

"Beauty knows no bounds": Exploring the influence of body size and content focus on women's body positivity

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ABSTRACT

Body positivity is a movement that promotes acceptance and appreciation of all body types. While previous research has explored the role of different model body types in promoting body positivity, there is no clear consensus on whether average-sized or plus-sized models would generate more positive outcomes. Moreover, most studies have focused on Western populations, leaving a gap in understanding within non-Western contexts. Average-sized models are often associated with assimilation effects, whereas plus-sized models are linked to downward comparison effects. Firstly, this study compared the influence of average-sized versus plus-sized models on body appreciation. Secondly, it also investigated how content focus—either aesthetic or functionality-based—influences body appreciation, and whether body size and content focus interact. Finally, the study explored the moderating role of self-esteem in shaping women's responses to model body size and content focus. Based on a 2 (average vs. plus-sized model) $\times$ 2 (aesthetic vs. functionality-focused content) experiment with 354 Chinese women ($M_{age} = 31.68$, $M_{BMI} = 21.46$), the findings showed that plus-sized models led to higher levels of body appreciation. Although content focus alone did not yield significant differences, an interaction effect emerged: plus-sized models paired with aesthetic-focused content led to greater body appreciation than the same content presented with average-sized models. Additionally, the study revealed a moderating effect of self-esteem on women's preferences for models and content. Women with high self-esteem preferred average-sized models and aesthetic-focused content, while those with low self-esteem favored plus-sized models and functionality-focused content. These findings have practical implications for promoting positive body image among women.

1. Introduction

Body image research has traditionally focused on women's negative experiences regarding appearance and body shape. Decades of empirical evidence have demonstrated that exposure to unrealistic body ideals, such as extremely thin or thin-toned figures, often leads to upward social comparisons and body dissatisfaction (Bessenoff, 2006; Tiggemann et al., 2009; Want, 2009). This dissatisfaction can drive harmful behaviors, including restrictive dieting, compulsive exercise, and the use of weight-loss products (Donovan et al., 2020; Sharpe et al., 2018), increasing the risk of eating disorders, depression, and other physical and mental health issues (Culbert et al., 2015; Gordon et al., 2017; Stice et al., 2000). Given these risks, identifying effective strategies to promote a healthier body image is essential.

In response, the body positivity movement has gained momentum on social media, challenging long-standing thin ideals. It encourages

individuals to accept and appreciate their appearance and functionality, regardless of body size or other physical features (Cohen et al., 2019; Cwynar-Horta, 2016; Tylka and Wood-Barcalow, 2015). On platforms like Instagram, larger-bodied women confidently display their bodies, often highlighting features such as cellulite, stomach rolls, stretch marks, and skin blemishes. These images are frequently accompanied by empowering captions like "fat girls can be active, fit and fabulous too!" or "all bodies are beautiful." (Cohen et al., 2019; Tiggemann et al., 2020). Exposure to such content has been shown to boost positive emotions, body satisfaction, and body appreciation (Clayton et al., 2017; Lou and Tse, 2021; Simon and Hurst, 2021; Tiggemann et al., 2020). As Smolak and Cash (2011) argued, advancing research on positive body image is vital for the field.

Similar movements have started to appear on Chinese social media, such as the hashtag "#Refusing Figure Anxiety". Lang et al. (2023) found that posts under this hashtag resemble those in the West, though the

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movement is still in its early stages. Traditional beauty norms continue to dominate, and plus-sized representation remains limited. For instance, Lang and colleagues reported that less than 10 % of relevant posts featured higher-weight individuals, compared to about two-thirds in Western contexts (Cohen et al., 2019). The cultural ideal of thinness has deep roots in China, reflected both in classical literature and contemporary media (Jackson et al., 2021; Leung et al., 2001). Research suggests that women in collectivist cultures like China are more vulnerable to social appearance pressures, leading to higher internalization of thin ideals and associated negative outcomes such as body dissatisfaction, body surveillance, and body shame (Jung and Forbes, 2007; Jung and Lee, 2006; Stojcic et al., 2020; Wu et al., 2023; Zhang, 2012). Given these cultural dynamics, there is an urgent need to promote body positivity in China to support more inclusive standards of beauty. Notably, research on body positivity within Chinese populations remains scarce, as most existing studies focus on Western samples (e.g., Cowles et al., 2023; Manning & Mulgrew, 2022; Vendemia & Robinson, 2022). As Andersen and Swami (2021) highlighted, a key limitation of body image research is its focus on Western, English-speaking populations. Therefore, we have expanded our exploration into the body image of the Chinese population.

Based on a 2 (body size: average vs. plus-sized model) $\times$ 2 (content focus: aesthetic vs. functionality-focused) online experiment, we investigated how different visual and textual cues affect body appreciation. We also explored whether self-esteem moderates these effects. Body appreciation, a core component of positive body image (Andrew et al., 2016; Tylka and Wood-Barcalow, 2015; Tylka, 2018), reflects respect for and acceptance of one's body and serves as a buffer against media-driven dissatisfaction (Andrew et al., 2015). Following prior research (e.g., Hepburn & Mulgrew, 2023; Tiggemann et al., 2020), we used body appreciation as the main outcome variable. By contrasting assimilation versus downward comparison effects and appearance-based versus function-based framing, this study aims to identify effective communication strategies for promoting body positivity and reducing body image concerns.

1.1. Body positivity and body size

The body positivity movement promotes acceptance and appreciation of diverse body sizes and appearances, challenging narrow and unrealistic beauty standards while promoting self-esteem and bodily autonomy (Cohen et al., 2019; Cwynar-Horta, 2016; Tylka and Wood-Barcalow, 2015). Originating from the 1960s fat acceptance movement in North America, it emerged as a feminist response to weight-based discrimination (Afful & Ricciardelli, 2015). A major turning point came in 2012 when model Tess Holliday launched the #effyourbeautystandards campaign on Instagram, directly confronting thin ideals. Since then, the body positivity movement has expanded (Cwynar-Horta, 2016). The growing prominence of this movement has fostered advocacy for positive body image (Cohen et al., 2019; Lazuka et al., 2020), stimulating increased scholarly attention within body image research. Positive body image represents the psychological outcome that the body positivity movement seeks to cultivate (Lazuka et al., 2020). While the movement operates collectively to resist societal beauty norms, positive body image is an individual attitude toward one's own body. It refers to respect and love for one's body, including its unique characteristics, functionality, and even features that diverge from sociocultural ideals (Mahlo and Tiggemann, 2016; Tylka and Wood-Barcalow, 2015; Wood-Barcalow et al., 2010). Positive body image is a multidimensional construct including six core components: body appreciation, body acceptance and love, broadly conceptualizing beauty, adaptive appearance investment, inner positivity, and filtering information in a body-protective manner (Tylka and Wood-Barcalow, 2015; Tylka, 2018).

Currently, some of the research on body positivity focuses on the impact of model size on women's positive body image. Consistent

evidence shows that exposure to thin-ideal models triggers upward social comparisons and lowers body satisfaction (de Valle et al., 2021). Conversely, both average- and plus-sized models have been found to foster more positive body image outcomes (Lou and Tse, 2021; Mulgrew et al., 2020; Vendemia and Robinson, 2022). Despite this progress, there is no clear consensus on which model size is more psychologically beneficial. For instance, Simon and Hurst (2021) reported greater emotional benefits from viewing average-sized models, while Clayton et al. (2017) found higher body satisfaction after exposure to plus-sized models. Meanwhile, Lou and Tse (2021) found no significant difference between the two. These inconsistencies underscore the need for further research to clarify which model size will lead to higher body appreciation.

1.2. Selective accessibility model and body size

Social comparison theory (SCT) (Festinger, 1954) has been widely applied in body image research to understand how women's self-perceptions shift after viewing model images. SCT posits that individuals are driven to evaluate themselves by comparing to others, typically using social standards as reference points (Festinger, 1954). However, these comparisons do not always produce the same outcomes. At times, individuals experience contrast—seeing themselves as different from the comparison standard; while at other times, they experience assimilation—seeing themselves as similar, even when the comparison standard remains the same. To explain this variability, Mussweiler (2001a), Mussweiler (2003) proposed selective accessibility model (SAM), an extension of SCT. SAM posits that comparison outcomes depend on whether individuals focus on similarities or differences relative to the target. When similarities are salient, individuals activate standard-consistent knowledge, leading to assimilation. Conversely, when differences are highlighted, contrast occurs, and individuals distance themselves from the comparison standard. Drawing from judgmental anchoring (Tversky & Kahneman, 1974), assimilation and contrast represent distinct responses to a given comparison standard—approaching or deviating from the standard—determined by the informational focus (Mussweiler and Strack, 1999; Mussweiler et al., 2004). While early SCT research emphasized the selection of comparison standards and their effects, SAM offers a more nuanced view by highlighting the cognitive processes shaping comparison direction (Mussweiler, 2003). Notably, SCT primarily emphasizes upward comparison to similar others for self-evaluation and group convergence (Festinger, 1954), which may not fully explain the dynamics of the body positivity movement, which aims at challenging rather than reinforcing thin-ideal norms. A large body of research has shown that exposure to the thin ideal often results in body dissatisfaction through upward comparison (Bessenoff, 2006; Tiggemann et al., 2009; Want, 2009). Therefore, we adopt SAM as a more comprehensive framework to explore how exposure to different body sizes affects body appreciation.

This study focuses on social comparisons concerning body size. Based on SAM, exposure to average-sized models may lead to assimilation, as many women perceive these bodies as more similar to their own (Lou & Tse, 2021). Numerous studies in social psychology suggest that individuals prefer others with similar appearances, attitudes, or personality traits (Montoya and Horton, 2013; Simons et al., 1970). Theories such as social identity theory (Tajfel & Turner, 1979) and similarity-attraction theory (Fiske, 2010) also support this effect. For instance, Bui (2017) found that individuals tend to favor celebrities of the same gender. Thus, under the process of assimilation, women may feel a stronger connection to average-sized models, reporting higher body appreciation through assimilation.

Complementing assimilation, SAM also accounts for contrast effects. Wills (1981) extended SCT by proposing that social comparisons are not only motivated by self-evaluation but also by self-enhancement, particularly under threat. Downward comparison occurs when individuals compare themselves to others perceived as worse off, leading

to more favorable self-evaluations (Gerber et al., 2018; Wills, 1981; Wills, 1987). Thus, women may feel better about themselves after seeing models who appear heavier than they are. Empirical research has shown that downward comparisons with plus-sized models can lead to a more positive body image (Clayton et al., 2017; Holmstrom, 2004; Vendemia and Robinson, 2022). In summary, we proposed a pair of competing hypotheses:

H1a. : According to the assimilation effect, women exposed to average-sized models will report higher levels of body appreciation compared to women exposed to plus-sized models;

H1b. : According to the downward comparison effect, women exposed to plus-sized models will report higher levels of body appreciation compared to women exposed to average-sized models.

1.3. Body conceptualization theory and content focus

Body conceptualization theory (Franzoi, 1995) divides the body into two aspects using a dichotomy: The "body-as-object" aspect focuses on the appearance and aesthetic qualities of body parts, whereas the "body-as-process" aspect emphasizes the overall functionality and capabilities of the body (Abbott & Barber, 2010). While the aesthetic/functionality distinction is theoretically useful, in practice, certain features may be perceived as both aesthetic and functional. For example, the "fit ideal" promoted on social media suggests that women should strive to be both lean and toned, merging physical functionality with aesthetic appeal through slogans like "strong is the new skinny" (Donovan et al., 2020).

Historically, body image research has prioritized appearance, often neglecting function (Linardon et al., 2023). However, recent scholarship argues for a more holistic view that includes both aspects (Mulgrew and Courtney, 2022; Mulgrew and Hennes, 2015). According to traditional gender norms, women are often socialized to value attractiveness, while men emphasize strength and capability (Franzoi, 1995). Consequently, body dissatisfaction in women frequently stems from concerns about weight and looks rather than functionality (McCabe et al., 2006). As an implicit quality, body functionality is not as easily objectified as appearance nor as readily subjected to social comparisons, because it typically does not constitute a key defining characteristic for women under traditional gender norms (Franzoi, 1995). When engaging in social comparisons, individuals predominantly evaluate themselves against standards on salient dimensions (Festinger, 1954; Mussweiler, 2001a). This inherent resistance to comparison positions functionality as a promising focal point for advancing body positivity.

In contrast, aesthetic-focused content may heighten self-objectification risk. According to objectification theory (Fredrickson & Roberts, 1997), repeated focus on appearance—both from others and oneself—leads to heightened body surveillance. Such content may draw attention to physical appearance and perpetuate beauty-based social comparisons. Critics argue that messages focusing on looks may unintentionally sustain patriarchal norms by continuing to anchor women's worth in their appearance (Darwin & Miller, 2021). For example, Cwynar-Horta (2016) found that the many body-positive Instagram posts still featured sexualized poses. And exposure to such images would increase both self- and other-objectification (Vendemia et al., 2021).

Functionality-focused messages, however, may offer a healthier alternative. Research shows that interventions emphasizing body functionality, such as structured writing tasks or positive reading materials, increased appearance satisfaction, body appreciation, and reduced self-objectification (Alleva et al., 2015; Mulgrew and Courtney, 2022). In contrast, even seemingly positive aesthetic-focused content (e.g., "We are all shaped in a way that is beautiful and unique") can inadvertently reinforce appearance-based standards, leading to upward comparisons and continued self-objectification (Brown and Tiggemann, 2020; Tiggemann et al., 2020). As Brathwaite et al. (2023) noted, appearance satisfaction can be fleeting and difficult to sustain, whereas functionality

promotes more enduring and inclusive body appreciation. Therefore, the following research hypothesis is proposed:

H2. : Women will demonstrate higher levels of body appreciation after reading functionality-focused content than aesthetic-focused content.

1.4. Body size and content focus

Social media content promoting body appreciation often relies on the integration of both visual and textual elements. However, existing research has largely overlooked the interaction between model body size and content focus. Preliminary evidence suggests that materials promote positive body image effectively when images and accompanying texts are congruent in their messaging (Hepburn and Mulgrew, 2023; Manning and Mulgrew, 2022). As suggested by the match-up hypothesis, there has to be a match between product and endorser in effective advertisements (Kamins, 1990; Wright, 2016). Therefore, when information promoting body positivity is coupled with idealized models, it does not promote a positive body image because models are perceived as dishonest (Tiggemann et al., 2020).

From the perspective of mainstream aesthetics, average-sized models are typically seen as more visually aligned with societal beauty norms compared to plus-sized models. As mentioned earlier, aesthetic-focused content often redirects attention to appearance and may inadvertently trigger upward comparisons (Brown and Tiggemann, 2020; Tiggemann et al., 2020). Therefore, the combination of average-sized models and aesthetic-focused content is more likely to result in scrutiny of the average-sized model's appearance (Tiggemann & Zinoviev, 2019), which can lead to greater body dissatisfaction. In contrast, plus-sized models, when paired with aesthetic-focused content, may evoke less upward comparison due to their deviation from the thin-ideal, thus potentially reducing appearance-based body dissatisfaction. However, the dynamics shift when the focus of the content is on functionality. For content focused on functionality, it may be more effective paired with models of average size, due to prevailing weight stigma and public assumptions that higher body weight is associated with poorer physical health (Puhl & Heuer, 2010). Therefore, when plus-sized models advocate for the functionality of fat, viewers may perceive them as less credible or even self-deceptive. Hence, the following research hypotheses are proposed:

H3a. The combination of aesthetic-focused content with plus-sized models will lead to higher levels of body appreciation than its combination with average-sized models;

H3b. The combination of functionality-focused content with average-sized models will lead to higher levels of body appreciation than its combination with plus-sized models.

1.5. Self-esteem, social comparison, and self-verification

Self-esteem, defined as an individual's overall sense of self-worth (Rosenberg, 1965), plays a key role in shaping body image (Grossbard et al., 2009) and social comparisons (Wills, 1987). According to downward comparison theory (Wills, 1981; Wills, 1987), individuals with low self-esteem are more likely to engage in downward comparisons for self-enhancement. In contrast, those with high self-esteem are more drawn to similar others (Griffitt, 1966; Leonard, 1975). Experimental evidence suggests that individuals with low self-esteem may experience positive emotional responses from downward comparisons, while those with high self-esteem do not necessarily feel better (Gibbons & McCoy, 1991). Applied to body image, women with low self-esteem may appreciate their bodies more after viewing plus-sized models, as these comparisons promote favorable self-reappraisal. Conversely, women with high self-esteem may prefer average-sized models who resemble themselves, reinforcing body satisfaction through similarity-based assimilation. Therefore, the following hypotheses are proposed:

H4a. Compared to women with low self-esteem, women with high self-esteem will demonstrate higher levels of body appreciation after exposure to average-sized models.

H4b. Compared to women with high self-esteem, women with low self-esteem will demonstrate higher levels of body appreciation after exposure to plus-sized models.

Beyond social comparison, self-verification theory (Swann et al., 1992) further explains how self-esteem influences the reception of body-positive messaging. According to this theory, individuals accept information aligned with their self-concept and reject inconsistent feedback. Thus, high self-esteem individuals, who typically hold a positive self-image (Murray et al., 2013; O'Dea and Abraham, 2000; Swami et al., 2016), are more likely to embrace aesthetic-focused body-positive messages (e.g., "Everyone is beautiful"). However, individuals with low self-esteem often experience these messages as incongruent with their self-view, leading to emotional discomfort or resistance. Research supports this perspective: Eisenstadt and Leippe (1994) noted that overly positive messages can amplify discrepancies between actual and ideal selves, worsening self-perception. Similarly, Wood et al. (2009) found that positive affirmations, such as "I'm a lovable person," can backfire for people with low self-esteem, triggering negative emotional reactions. In such cases, self-verification motives prompt message rejection. Compared with aesthetic-focused content that carries a subjective evaluative tone for one's appearance, functionality-focused content emphasizes bodily abilities (e.g., "My legs carry me through life"). Such messages are more closely tied to lived, everyday experiences, making them easier for individuals with low self-esteem to accept. For instance, while someone may hesitate to agree that a chubby belly is beautiful according to her personal aesthetic standards, she is more likely to resonate with reminders of her body's everyday usefulness, such as the support her legs provide for walking and running. Thus, we proposed:

H5a. : Compared to women with low self-esteem, women with high self-esteem will demonstrate higher levels of body appreciation after reading aesthetic-focused content;

H5b. : Compared to women with high self-esteem, women with low self-esteem will demonstrate higher levels of body appreciation after reading functionality-focused content.

2. Method

2.1. Participants

We distributed the questionnaire in March 2024 through the *Credamo* platform (<https://www.credamo.com/#/>), one of China's most widely used online data collection platforms. A total of 354 women participated in this experiment, aged between 18 and 53 years ($M = 31.68$, $SD = 6.72$), with an average BMI of 21.46 ($SD = 2.68$). Participants who passed the manipulation check were compensated with 3 CNY (approximately equivalent to 0.4 USD).

2.2. Design and procedure

This study employed a 2 (body size: average vs. plus-sized model) $\times$ 2 (content focus: aesthetic vs. functionality-focused content) between-subjects factorial experimental design. Participants were automatically and randomly assigned by the *Credamo* platform to one of four groups with approximately equal sizes. To conceal the experimental intent, participants were informed that they were invited to participate in a new media content browsing evaluation. After reading the experimental materials, they completed measurements for all variables. The entire process took approximately 7 min. This study received ethical approval from the internal review board of Shanghai Jiao Tong University (no. H202301641).

2.3. Randomization check

One-way analyses of variance on age and BMI were conducted to verify that random assignment to the four conditions was successful. The results showed that participants who were assigned to these four conditions did not differ significantly in age ($F(3, 350) = 0.67$, $p = .569$) or BMI ($F(3, 350) = 0.39$, $p = .760$).

2.4. Stimuli

Considering that the research aims to explore the effects of the body positivity campaign strategy, we designed the experimental materials as articles from a WeChat official account based on Chinese social media usage habits. We selected an ordinary woman (non-celebrity) with an average body size from *Xiaohongshu* (often referred to as the Chinese equivalent of Instagram) to serve as the model for the average-sized model group. To construct the plus-sized model condition, we digitally altered the same model to appear plus-sized. This modification involved enlarging not only her body but also her face to maintain proportionality. The Chinese beauty ideal typically emphasizes facial symmetry, defined eyebrows, large and bright eyes, a straight nose, a small cherry-shaped mouth, and a petite face with fair skin (Jackson and Chen, 2008; Jung, 2018; Lee and Lee, 1996). To promote diversity in body and appearance, the model selected did not fully conform to these criteria. For example, her mouth was relatively large, and her skin tone was a light tan, diverging from one of the most enduring beauty ideals in both traditional and contemporary Chinese society—fair skin (Jung, 2018).

Drawing from the previous study (Häfner, 2004), we crafted four article titles to align with the content of each experimental condition: "To all the unique/ordinary girls out there: Be yourself, that is your beauty/strength." According to the similarity/dissimilarity priming mechanism proposed in SAM (Mussweiler et al., 2004; Mussweiler, 2001b), similarity priming leads to assimilation effects, while dissimilarity priming produces contrast effects. Accordingly, the terms "unique" and "ordinary" were used to correspond to the plus-sized and average-sized model groups, respectively, aiming to elicit contrast or assimilation effects. "Unique" was intended to highlight dissimilarity, whereas "ordinary" was used to foster a sense of similarity (Häfner, 2004). Meanwhile, the closing terms "beauty" and "strength" reflected the aesthetic-focused and functionality-focused content conditions. This framing strategy was intended to reinforce the message of the accompanying image and textual content.

The articles were narrated in the first-person perspective and consisted of two main parts, presented in written text form. The first part involved the model describing her own body. For instance, the plus-sized model informed participants that her BMI was as high as 35, while the average-sized model mentioned that her BMI fell within the normal range. The second part of the article discussed either aesthetic or functionality-focused content. Both types of messages were delivered in a positive tone to avoid potential confounding effects and to remain consistent with the orientation of body positivity. Aesthetic-focused content emphasized self-love and acceptance of bodily imperfections, such as "It is the fat on the body that gives women their unique curve beauty." Functionality-focused content highlighted physical capabilities and health benefits (Alleva et al., 2015), such as "My feet are faithful companions that carry me through wilderness paths" or "Low body fat may lead to functional hypothalamic amenorrhea, which in turn can cause osteoporosis." Each article consisted of approximately 400 words and one photo of the model. Participants were randomly assigned to read one article. All the stimuli above could be found in the supplementary file.

2.5. Manipulation check

The manipulation check was conducted using two items. The first item required participants to indicate whether the model had a thin/

average/plus-sized body. The second item asked participants to identify the primary emphasis of the text, whether it emphasized "appreciation of physical appearance and beauty" or "appreciation of the body's functional value," corresponding to aesthetic/functionality-focused content, respectively. Failure in the manipulation check resulted in exclusion from subsequent data analyses.

We assessed perceived similarity using a five-point Likert scale, asking participants to rate their similarity to the model in terms of body shape, body size, and weight (1 = *not similar at all*, 5 = *very similar*), based on the method adapted from Lou and Tse (2021). An independent samples *t*-test revealed that the average-sized model ($M = 3.59$, $SD = 1.05$) elicited a significantly higher level of perceived similarity ($t(352) = 11.71$, $p < .001$) than the plus-sized model ($M = 2.25$, $SD = 1.11$), confirming that the manipulation of body size was successful.

2.6. Measures

2.6.1. Body appreciation

We utilized the Chinese version of the Body Appreciation Scale-2, adapted by Ma et al. (2022). Examples included "I respect my body," "I feel good about my body," and "I take a positive attitude towards my body." Item 3 and Item 5 were deleted from the scale due to factor loadings below 0.3, resulting in body appreciation being measured by eight items in this study. Participants rated these items on a 5-point Likert-type scale, ranging from 1 (*completely disagree*) to 5 (*completely agree*) ($M = 4.33$, $SD = 0.37$, Cronbach's $\alpha = .70$, $\chi^2 = 71.252$, $\chi^2/df = 3.563$, $p < .001$, CFI = .855, RMSEA = .085 (90 % CI [.064, .107]), SRMR = .056).

2.6.2. Self-esteem

We employed a Chinese version of the scale adapted from the Rosenberg Self-Esteem Scale (RSES) (Rosenberg, 1965), which consists of 10 items (Ji & Yu, 1999). Participants rated these items on a 4-point Likert-type scale, ranging from 1 (*completely mismatched*) to 4 (*completely matching*) ($M = 3.40$, $SD = 0.36$, Cronbach's $\alpha = .75$).

To avoid multicollinearity issues, the moderator variable "self-esteem" was centered following the recommendation by Aiken and West (1991). Collinearity diagnostics were conducted due to the correlation among variables. With variance inflation factors values for all variables found to be less than 10, our analysis was not affected by significant multicollinearity issues (O'Brien, 2007).

Table 1
Parameter estimates for the multivariate regressions.

Predictors	Body appreciation <i>B</i>	<i>SE</i>
Intercept	4.428***	0.143
Body size (reference = average-sized model)	0.103*	0.047
Content focus (reference = aesthetic-focused content)	0.074	0.049
Self-esteem	0.774***	0.079
Age	0.001	0.003
BMI	-0.009	0.007
Body size: content focus	-0.177**	0.067
Content focus: self-esteem	-0.210*	0.095
Body size: self-esteem	-0.328***	0.095
Observations	354	
R^2	.298	
Adjusted R^2	.282	
<i>F</i>	18.304*** ($df = 8; 345$)	

Note.

* $p < .05$.

** $p < .01$.

*** $p < .001$.

3. Results

The results of the multivariate regression model (see Table 1) revealed a significant impact of body size on body appreciation ($B = 0.10$, $p = .028$). The plus-sized model induced higher body appreciation levels compared to the average-sized model, thus H1b was supported. Content focus did not significantly affect body appreciation ($B = 0.07$, $p = .132$), failing to support H2.

There was an interaction effect between body size and content focus ($B = -0.18$, $p = .009$). Simple slopes analysis with content focus as the moderator established that the slope for the functionality-focused content was not statistically significant ($B = -0.07$, $SE = 0.05$, $t(345) = -1.53$, $p = .126$), failing to support H3b; however, the slope for aesthetic-focused content was statistically significant ($B = 0.10$, $SE = 0.05$, $t(345) = 2.21$, $p = .028$), thus H3a was supported. The interaction plot (see Fig. 1) illustrates that aesthetic-focused content narrated by a plus-sized model led to higher body appreciation than content narrated by an average-sized model. However, there was no difference in impact for functionality-focused content, regardless of whether it was narrated by a plus-sized or average-sized model.

Self-esteem significantly moderated the relationship between body size and body appreciation statistically ($B = -0.33$, $p < .001$). Simple slopes analysis with body size as the moderator established that the slope for the average-sized model was statistically significant ($B = 0.77$, $SE = 0.08$, $t(345) = 9.78$, $p < .001$); and the slope for the plus-sized model was also statistically significant ($B = 0.45$, $SE = 0.08$, $t(345) = 5.63$, $p < .001$). The moderation plot (see Fig. 2) illustrates that compared to women low in self-esteem, women high in self-esteem exhibited higher levels of body appreciation after exposure to an average-sized model; conversely, compared to women high in self-esteem, women low in self-esteem demonstrated higher levels of body appreciation after exposure to a plus-sized model. H4a and H4b were supported.

Self-esteem significantly moderated the relationship between content focus and body appreciation statistically ($B = -0.21$, $p = .028$). Simple slopes analysis with content focus as the moderator established that the slope for the aesthetic-focused content was statistically significant ($B = 0.77$, $SE = 0.08$, $t(345) = 9.78$, $p < .001$); and the slope for functionality-focused content was also statistically significant ($B = 0.56$, $SE = 0.09$, $t(345) = 6.18$, $p < .001$). The moderation plot (see Fig. 3) illustrates that compared to women low in self-esteem, women high in self-esteem exhibited higher levels of body appreciation after reading aesthetic-focused content; conversely, compared to women high in self-esteem, women low in self-esteem demonstrated higher levels of body appreciation after reading functionality-focused content. H5a and H5b were supported.

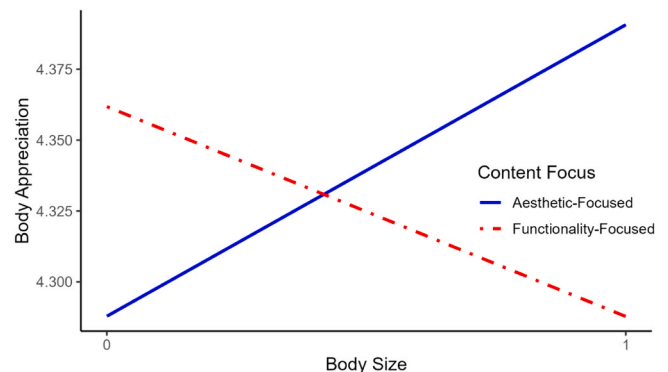


Fig. 1. Interaction effects between body size and content focus. Note. Body Size: 0 = average-sized model, 1 = plus-sized model.

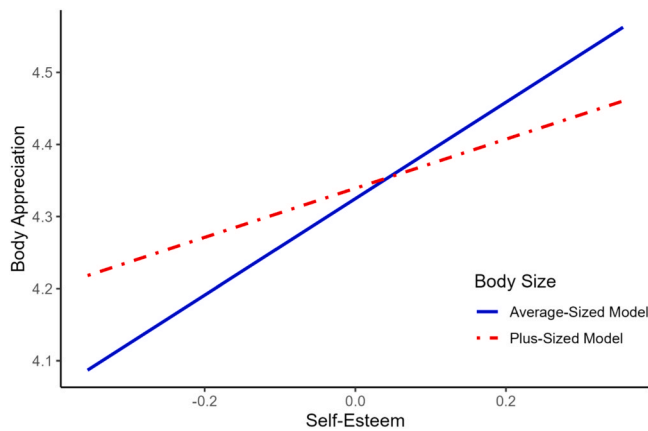


Fig. 2. Moderating role of self-esteem in the relationship between body size and body appreciation.

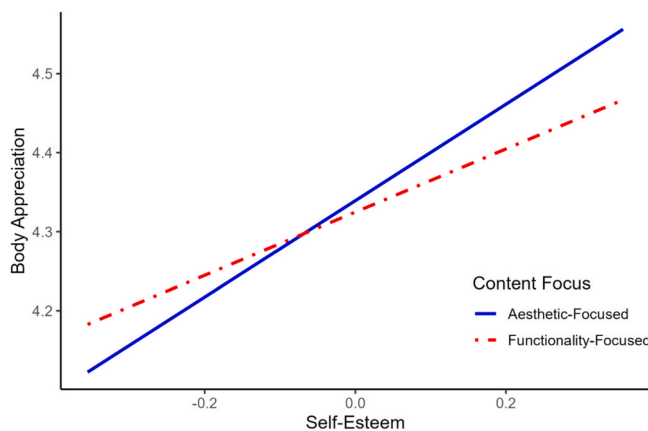


Fig. 3. Moderating role of self-esteem in the relationship between content focus and body appreciation.

4. Discussion

4.1. Key findings

To promote women's body appreciation, this study investigated the communication effects of potential strategies in body positivity campaigns, focusing on two aspects: model body size and content focus. Specifically, we found that plus-sized models led to higher levels of body appreciation compared to average-sized models. Furthermore, there was no statistically significant difference between aesthetic and functionality-focused content. However, an interaction effect was observed between body size and content focus, indicating that aesthetic-focused content moderated the relationship between body size and body appreciation. Additionally, we examined the moderation effects of self-esteem on the relationships among body size, content focus, and body appreciation. The results showed that average-sized models and aesthetic-focused content led to higher levels of body appreciation among women with high self-esteem. In contrast, plus-sized models and functionality-focused content generated greater body appreciation for women with low self-esteem.

Our findings indicated that plus-sized models were linked with higher body appreciation levels than average-sized models, consistent with the results of Clayton et al. (2017). While earlier studies have not clearly identified whether plus-sized or average-sized models are more psychologically beneficial, our research provides further insight into this question. Therefore, based on the finding, we assume that downward comparison effects may lead to stronger body appreciation than

assimilation effects in mitigating women's negative body image issues. Overall, our study suggests that individuals benefit more from comparing themselves to less idealized standards, which can lead to greater body appreciation. Body dissatisfaction often stems from the gap between one's actual and ideal body (Bessenoff, 2006; Halliwell and Dittmar, 2006; Kakar et al., 2023a), a concept grounded in self-discrepancy theory (Higgins, 1987). When women perceive themselves as falling short of the societal ideal, negative emotions like anxiety and sadness may emerge. Introducing plus-sized models as attractive representations challenges the thin-ideal and narrows this discrepancy. As women see their bodies more positively, especially when they equal or surpass the comparison standard, body appreciation increases.

In addition, the reason why plus-sized models are accompanied by greater body appreciation among Chinese women may lie in cultural factors. Within the context of collectivism, which emphasizes conformity and adherence to social norms (Triandis, 1989), body appreciation may be more closely linked to social acceptance than to individualistic self-affirmation. Empirical research has shown that Chinese women report greater pressure from family, peers, and media to conform to societal beauty ideals compared to Croatian women (Stojic et al., 2020). In collectivist cultures, women often monitor and regulate their bodies to ensure they align with accepted norms, especially the prevailing ideal of thinness (Jung and Lee, 2006; Kakar et al., 2023b). Therefore, viewing a plus-sized figure that challenges the dominant thin ideal may help these women recognize alternative standards of beauty, potentially liberating them from restrictive cultural expectations. Until plus-sized representations become mainstream and gain broader societal recognition, body dissatisfaction among Chinese women is likely to persist. Furthermore, SCT suggests that the outcomes of comparison are shaped by self-motives (Wood, 1989). Individuals driven by self-improvement motives are more likely to assimilate with upward comparison targets, whereas those driven by self-enhancement motives tend to contrast with downward targets, especially under high pressure or threat. Given the persistent appearance-related pressure in collectivist societies, Chinese women may be more inclined toward self-enhancement, making downward comparisons (e.g., with plus-sized models) more psychologically beneficial.

Although plus-sized models promote higher body appreciation than average-sized models do, this effect varies among individuals with different levels of self-esteem. Following downward comparison theory (Wills, 1981; Wills, 1987) and supporting studies (Gibbons and McCoy, 1991; Griffitt, 1966; Leonard, 1975), women with low self-esteem experience greater body appreciation when exposed to plus-sized models. This is likely because they prefer comparisons that position them more favorably. Conversely, women with high self-esteem demonstrate higher body appreciation after exposure to average-sized models, as they are more inclined to assimilate with similar other.

Contrary to H2, we found no difference between aesthetic and functionality-focused content in promoting body appreciation. As highlighted by some studies, "A picture is worth a thousand words" (Brown and Tiggemann, 2020; Hepburn and Mulgrew, 2023; Tiggemann et al., 2020). During rapid browsing, visual images are more attention-grabbing than accompanying verbal text. Participants' responses were largely shaped by the models' appearance, overshadowing the effects of textual content. A similar study reached a comparable conclusion: neither body appreciation, body functionality, nor a combination message of the two impacted body image. The type of positive information was not significant, as the influence of experimental materials on body image was primarily driven by the model's physique (Hepburn & Mulgrew, 2023). Future studies could use the eye-tracking technology to test this assumption that whether the non-effectiveness of text owe to the visual domination. Additionally, our single-exposure design warrants investigation into whether amplified stimulus exposure intensity moderates these effects.

Despite the lack of main effects of content focus, it showed

differences in promoting body appreciation after being moderated by self-esteem. Women with higher self-esteem, who tend to feel more confident about their appearance, responded more positively to aesthetic-focused content. This is consistent with the self-verification motive, which suggests that individuals seek feedback that aligns with their self-perception. Conversely, aesthetic-focused content exacerbates the gap between ideal and actual selves for women with lower self-esteem, who are generally less satisfied with their bodies and appearance, making it challenging to promote body appreciation. For them, functionality-focused content proved more beneficial. Emphasizing bodily capabilities (e.g., "My feet carry me through wilderness paths") highlighted aspects of the body that are grounded in daily experience, which could be more readily accepted than subjective praise of appearance.

In addition, we found an interaction between body size and content focus, with the combination of aesthetic-focused content and a plus-sized model achieving the highest body appreciation levels. While few studies have examined the interaction between model body size and information type, [Hepburn and Mulgrew \(2023\)](#) explored a related interaction between image type and message type but found no significant effect. This discrepancy may stem from differences in experimental materials. Their study used short phrases overlaid on images, which may have had limited impact due to brevity and reduced cognitive engagement. In contrast, our materials included a 400-word article, allowing participants more time for reflection and processing, which may have strengthened the effects. This interaction underscores the importance of who delivers the aesthetic-focused message. Consistent with previous research findings ([Brown and Tiggemann, 2020](#); [Tiggemann et al., 2020](#)), pairing aesthetic-focused content with an average-sized model drew participants' attention to the appearance of the average-sized model and led to upward comparisons. However, the situation differs when aesthetic-focused content is delivered by a plus-sized model, who is further away from the thin-ideal. Firstly, the plus-sized model is less likely to evoke upward comparisons. Even if she delivers aesthetic-focused content, women are less likely to feel dissatisfied with their bodies compared to when they see an average-sized model. Secondly, research suggests that the best outcomes are achieved with consistency between images and text ([Hepburn and Mulgrew, 2023](#); [Manning and Mulgrew, 2022](#)), aligned with the match-up hypothesis. In this case, aesthetic-focused content aligns more naturally with a plus-sized model, reinforcing the message that beauty is not confined to a single body type. This consistency communicates the idea that "beauty knows no bounds," helping to normalize a broader range of body ideals. Additionally, a recent study also explored the interaction between model body size and content focus, but reported findings that diverge from ours ([Kharkwal et al., 2025](#)). Specifically, Instagram gym advertisements featuring plus-sized models paired with health-and-wellness slogans (as opposed to physique-focused slogans) led to greater body satisfaction, although the interaction effect was non-significant. These inconsistent results may be explained by cultural differences in self-construal between Western and Eastern populations ([Markus & Kitayama, 1991](#)). In collectivist cultures such as China, where individuals tend to hold an interdependent self-construal, women are more likely to prioritize social expectations and others' evaluations. Therefore, body positivity messages may lead to more positive outcomes when they signal that chubbiness has achieved broader social acceptance and legitimacy. Conversely, Western populations' independent self-construal emphasizes internal values and abilities, making functional messaging with plus-sized models more resonant.

4.2. Strengths and limitations of the study

The study makes two main contributions. Firstly, it compared the influence of average and plus-sized models in promoting body appreciation, revealing that plus-sized models were associated with higher body appreciation levels. This adds empirical evidence to the existing

literature on this topic. Moreover, we discovered that the effects of model body size varied depending on self-esteem levels, highlighting the role of individual psychological factors. Secondly, this study is one of the few to explore the interaction between model body size and content type, finding a notable interaction between the two factors.

The findings have practical implications for guiding body positivity campaigns. Because self-esteem influences preferences for body size, campaigns are encouraged to feature a broad spectrum of body types, including but not limited to plus-sized models, especially when presenting aesthetically focused messages. This strategy aligns with the "Beauty Beyond Size" concept, which advocates for diverse representations of beauty. For example, collaborating with influencers of varying body sizes to create visually compelling content, such as bold fashion editorials or empowering photo series, can highlight beauty beyond traditional norms. Furthermore, this principle extends to message design, which may benefit from combining aesthetic and functionality-focused elements to engage a broader audience. For instance, a campaign could begin by celebrating the beauty of a rounded belly and then highlight its functional role in providing warmth.

Despite its contributions, this study has several limitations that warrant consideration and suggest directions for future research. Firstly, we acknowledge methodological limitations. A state version of the Body Appreciation Scale-2 ([Homan, 2016](#)) would have been more appropriate than the trait version used, particularly in studies examining immediate responses to media exposure. As [Thompson \(2004\)](#) suggested, state measures may be more sensitive to short-term effects triggered by experimental stimuli. Nonetheless, given that body image is not entirely stable and can be shaped by experiential and contextual factors ([Melnik et al., 2004](#); [Wood-Barcalow et al., 2010](#)), our study along with others using trait measures (e.g., [Alleva et al., 2015](#); [Alleva et al., 2020](#)) still detected meaningful short-term effects. In addition, the Body Appreciation Scale-2 used in the study is a newly adapted Chinese version ([Ma et al., 2022](#)), which has not yet been widely validated in Chinese culture. Cultural and linguistic differences between Chinese and English may still need to be addressed. Moreover, although our post-test only design helps reduce demand characteristics ([Krawczyk et al., 2012](#)), a pretest-posttest design would allow researchers to account for baseline differences in body appreciation, offering a clearer view of change and improving internal validity. Secondly, this study adopted a dichotomous or approximate dichotomous approach to body size and content focus. Future research could investigate the mixed effects of different body sizes and content focus types by presenting models of various sizes and combining both aesthetic and functionality-focused content. Although we implemented several measures to strengthen the body size manipulation, including random assignment, exclusion of participants who failed to recognize the model's body size, and a manipulation check, we recommend pretesting images across conditions to ensure consistent perceptual validity and enhance result robustness. It is also worth noting that, as a limitation, the stimuli did not encompass all components of body functionality, focusing primarily on functionality related to body fatness. Future studies could incorporate a broader range of body functionalities. Thirdly, our use of WeChat article stimuli may not fully capture the complexity of the broader social media environment, which often includes likes, comments, and rapid information flow. Thus, the external validity of our findings is limited. Future research should examine whether the format of media—such as posters, websites, or videos—modulates the effectiveness of body-positive messaging. Additionally, using a single image and article as stimuli may limit generalizability. Future studies should include multiple images featuring different faces to minimize confounding effects and strengthen robustness. Lastly, while current body positivity research has largely focused on women, there is a growing need to investigate negative body image among men, who are often influenced by the muscular ideal promoted in the media ([Dondzilo et al., 2019](#); [Galioto and Crowther, 2013](#)). Future research could expand the scope of body positivity research and theory to include men.

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CRediT authorship contribution statement

Weixi Zeng: Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Junyi Chen:** Writing – review & editing, Validation, Methodology, Investigation, Conceptualization. **Yi-Xin Zhou:** Writing – review & editing, Validation, Supervision, Software, Methodology. **Nainan Wen:** Validation, Supervision, Resources, Methodology, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.bodyim.2025.101985](https://doi.org/10.1016/j.bodyim.2025.101985).

Data Availability

The data supporting this study's findings are publicly available at OSF: <https://osf.io/jgp29/>.

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